

GEORGE LARSON

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SUMMARY

Updated 2026-06 Director of Technology with 25+ years building and operating software, infrastructure, and manufacturing systems. I have led teams at Sony Electronics, scaled a medical billing startup to national reach, programmed PLCs on production lines, and directed technology for a mid-size manufacturer. I do Linux platforms, DevOps automation, ecommerce, and security programs. I stay close to the keyboard even when the job is strategic. I build AI tooling that actually ships: multi-agent infrastructure, MCP servers, model-routing pipelines, SDKs in the Venice AI ecosystem, and a personal review methodology now landing surgical fixes upstream in mitmproxy, n8n, ComfyUI, and Charm's Crush.

RECOGNITION

- Top 1% of TypeScript engineers globally (Algora, 1,200+ contributions over the past year).
- KEY METRICS (FM Expressions, at separation April 2026)
- 36 public assets scanned continuously via HostedScan + OWASP ZAP with 33 findings closed in the past 60 days and zero critical/high issues in backlog.
 - 36 active GitLab projects (11 users, 2 groups), 2,204 merge requests, 739 CI pipelines (avg 565 successes / 58 fails) and 16 managed SSH keys.
 - Mattermost workspace (retired 2026) with 7 teams, 40 channels, ~1.0M lifetime posts, 313 calls, and alert channels wired to uptime, upgrade, helpdesk, GitLab, and web-critical events.
 - Zammad helpdesk processed 1,487 tickets YTD (~150–206/month in Jun–Aug) with ≥97% close rate, 68% resolved inside 24 hours (56% inside 8 hours), zero reopen events.
 - Uptime Kuma monitors 31 services (GitLab, Helpdesk, Shopify APIs, FTP, chat, docs, etc.) as part of 24×7 observability feeding alert channels.

CURRENT FOCUS

- Independent AI-systems R&D: a four-node Tailscale mesh, ten coding-agent CLIs (Claude Code, opencode, gemini-cli, Codex CLI, Kimi Code, OpenHands, Aider, Octofriend, Crush, letta-code), and tiered inference across thirteen authenticated providers routed for cost, latency, and capability.
- Shipping surgical bug fixes upstream as public-record evidence of the operational style: mitmproxy, n8n, charmbracelet/crush, ComfyUI, and the IronClaw agent platform.
- Building in the Venice AI SDK ecosystem (TypeScript SDK, PHP SDK, deep-research-privacy at 36★) with continued systems work in Rust, Go, and Zig (netgrep, termshark, wiregraph, rust-connect).
- Strengthening security programs, compliance boundaries, and knowledge management systems that keep data and people protected.

CORE STRENGTHS

Software Development and Engineering

- Full stack delivery across PHP, Perl, SQL, Python, Go, Rust, C#, TypeScript, C++. PHP and Perl have been the career staples; the rest sit alongside.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.

DevOps and Infrastructure Automation

- CI and CD design, infrastructure as code, container orchestration, and repeatable releases on self-hosted runners.
- Observability, incident response, high-availability design (replication, backups, CDN), and pragmatic SLO tracking for hybrid environments.

Technology Leadership and Strategy

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams.

Security and Privacy

- Network hardening, IAM, HostedScan/ZAP/Nikto assessment loops, and zero-trust rollouts.
- Regulated-data guardrails and tabletop response drills.

Data Engineering and Automation

- OCR digitization, ETL pipelines, semantic search, and knowledge-base design.
- Workflow automation using scripting, open models, and RAG pipelines where the payoff is real.

OPEN SOURCE PROJECTS

Recent Upstream Contributions (April-May 2026)

- mitmproxy/mitmproxy #8199 (+23/-0): Flow.kill() resume-event fix so proxy teardown closes the client connection (PR open).
- mitmproxy/mitmproxy #8200: KillInjected event so Flow.kill() actually disconnects in-transit flows (PR open).
- n8n-io/n8n #29516 (+288/-2 across 6 files, 48 jest tests passing): per-runner consecutive-timeout counter for stuck task-runner channel detection and recovery; reduced detection latency from hours to ~6s. Maintainer-accepted into Linear (PR open).
- charmbracelet/crush #2749 (+50) and #2750 (+180): LSP score-based client selection and SingleFileSupport handling so single-file servers start without root markers (PRs open, both labeled area: lsp).
- Comfy-Org/ComfyUI_frontend #11682 (graph-traversal cycle guards) and #11686 (keybindings respecting Reka DismissibleLayer): both checks-passing, standing.
- nearai/ironclaw #2390: vision-detail default fix for OpenAI-compatible providers across the agent platform.

Venice AI SDKs (2025 to Present)

- deep-research-privacy (36 stars, TypeScript): privacy-first AI research tool with an iterative search-and-synthesis pipeline, plus published TypeScript and PHP SDKs (venice-dev-tools, venice-ai-php) for the Venice API.

LLM Showdown (2026 to Present)

- llm-showdown-data. Companion dataset for the blog series benchmarking 15+ free coding models on a \$25 VPS using opencode and a ralph-wiggum-style iterative loop.
- Designed and runs the evaluation rig; publishes per-round results comparing model behavior on real coding tasks.

Fracture (2 stars, TypeScript), Modernization Case Study (2026)

- Modernizes Mozilla's 2012 BrowserQuest into a contemporary multiplayer architecture with eight integrated AI services: RAG, semantic search, agent-driven triage, and content moderation.
 - Live demo at fracture.georgelarsen.me. The case study itself is the deliverable: how to take a frozen reference codebase from a major vendor and bring it forward without rewriting from scratch.
- nullclaw + ironclaw, Agent Mesh (Zig + Rust, 2026)
- nullclaw: public-facing AI concierge ("Nully") on a hardened perimeter box. Tiered inference (Haiku for conversational triage, Sonnet for code analysis). Ran a full self-hosted comms stack (Ergo IRC, gamja web client, Cloudflare-proxied WebSocket, Let's Encrypt TLS) before transitioning to a static-site origin.
 - ironclaw: private personal agent on a separate system with distinct security boundaries. Agent-to-agent routing between the two.
 - Extended into a multi-node Agent Zero mesh with A2A routing, dedicated chat and utility models, and per-node project isolation.
- netgrep (Rust), Creator and Maintainer (2026 to Present)
- Modern replacement for ngrep: live packet capture and pcap analysis with TCP stream reassembly, TLS 1.2/1.3 decryption, HTTP/2 and DNS parsing, and an interactive terminal UI.
 - Maintains ngrep-compatible CLI flags; outputs colored text, JSON, hex dump, or pcap.
- termshark (Go), Contributor and Maintainer (2026 to Present)
- Modernized fork of termshark, a terminal UI for Wireshark/tshark.
 - Refactored architecture, improved test coverage, updated to Go 1.22+ idioms, and added an experimental browser-based web interface.
- wiregraph (Rust, 2026 to Present)
- GPU-rendered network traffic visualizer. Hosts as nodes, flows as edges, real-time topology view of a live capture session.

WORK EXPERIENCE

Director of Technology, FM Expressions (2018 to April 2026)

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels.
- Absorbed the infrastructure team and its full stack when the role was eliminated during a downturn, running technology and infrastructure together with no added headcount.
- Shipped production AI inside the business from 2023: a customer-facing DALL-E design assistant for custom heat transfers, an OpenAI/Pythagora code-generation tool on MongoDB, an AI knowledge platform (wiki.fmx.ink), a customer-service AI/BI integration, and a Python/FastAPI templating system bridging the Fanatics and VivX vendor pipelines.
- Improved database performance by 93 percent while driving critical systems to 99 percent uptime; cut MySQL slow-query volume by 96 percent to accelerate incident response.
- Built a knowledge management platform (Wiki.js, Gitea, Qdrant, ETL pipelines) that captures production logs, transcripts, and operator notes so teams can search, surface, and act on institutional knowledge in real time.
- Resolved conflicting manufacturing documentation, created a style guide and glossary, and instituted monthly and quarterly review cycles to keep safety and operations guidance consistent.
- Reduced a 1.03 GB legacy software repository (4,656 binary artifacts) to a clean, LFS-ready codebase, eliminating clone failures and enabling sustainable development practices.
- Hardened hybrid infrastructure with Authentik SSO, HashiCorp Vault, automated TLS renewal, firewall hardening (iptables, UFW, CrowdSec, fail2ban), DMARC/SPF/DKIM enforcement, HostedScan/OWASP ZAP sweeps across 36 public assets, and Red Sift OnDMARC (p=reject, 1.5M emails/month at 99.48% compliance) to keep critical/high findings at zero; closed 33 issues in the 60 days prior to separation.
- Ran the security program's people-and-email layer: Mimecast email security, a KnowBe4 security-awareness and phishing-simulation program, and live response to an active production compromise.
- Ran GitLab (36 projects, 11 users, 739 CI pipelines) and Mattermost (7 teams, 40 channels, ~1M posts, retired 2026) as the collaboration core tied into Redmine (22 cross-functional projects), Zammad (~200 tickets/month with 68% closed inside 24 hours), and Uptime Kuma (31 monitors feeding alert channels).
- Maintained high-availability platforms across a heterogeneous Windows-and-Linux estate: a high-availability MySQL cluster and MSSQL, VMware ESXi with Datto backups, on-prem Active Directory with Entra ID hybrid identity (Group Policy, PowerShell), multi-vendor firewalls (Cisco, SonicWall, pfSense), Cisco Meraki switching, OpenVPN, and a staged CDN migration from self-hosted edges to Contabo and Cloudflare; led countless OS and database migrations.
- Built the Mashy internal dashboard plus tooling such as Psono vault, PrivateBin, phpMyAdmin/Adminer, DBeaver, knowledge mining utilities, and status pages so teams access credentials, data stores, and observability in one click.
- Drove adoption metrics for the knowledge stack to quantify time-to-answer gains for production and customer operations.
- Captured Meteor print-head PCAPs and tuned Nikto, Metasploit, and custom fuzz rigs to secure industrial protocols while leading the move from VPN toward zero-trust access.
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.
- Built and operated integrations across the print and fulfillment vendor stack (Pace/EFI, Printronix, VivX, Fanatics, Alphabroder, SignTronics) plus ecommerce platforms (Shopify, Magento) and marketing tooling (HubSpot, Power BI), so the production floor, order pipeline, and analytics layer share a coherent data model.
- Administered the prepress production stack (Esko, Prinergy, RIP workflows) and 3PL logistics integrations that kept the manufacturing floor running.
- Built financial-modeling tools using AI/ML to analyze company performance and forecast scenarios; presented findings directly to executive leadership to inform strategic decisions during a difficult business period.

Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)

- Helped scale the company from a 50 person startup to a national leader in medical billing (Inc 5000, ranked #34 in Health and #503 overall).
- Designed the EDI framework, ACH transmission, ERP integration, payer portal, Gold Standard Review, and recovery analytics layers of the platform.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead, HIPAA/HITECH SME, and Linux SME for the medical billing platform; migrated project management to Redmine with custom plugins.
- Deployed VMware ESXi covering web, database, DHCP, DNS, and ERP servers across the platform.
- Built OCR bill entry validation that reduced errors and sped turnaround.

Software Engineer and Team Manager, Sony Electronics (2004 to 2009)

- Joined a service operation that was effectively bare shelves at start; directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center from there.

- Designed and ran the TiVo firmware upgrade station for the United States market.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Created an AS/400 parts ordering interface that saved millions in operational spend.
- Front-end software work delivered roughly 30 percent productivity gain and 12 percent expense reduction.
- Managed more than 50 combined technical and non-technical staff, including roughly 30 contracted personnel.
- Developed a VT-52 terminal emulation system used for inventory management.

Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)

- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and Adept V++ robotics in-house, tuned sensors, actuators, and industrial controls on the production line, and kept robotics development internal to limit vendor cost.

Earlier Career Highlights (1990s to 2000s)

- Software developer and webmaster for non-profit organizations.
- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS and kept one of the first internet-connected BBS systems alive.
- Contract developer for Daifuku USA working on material handling support systems.
- Paid programming and database work 1994 to 2000 via Pacific Judgment Recovery (Brice, OH), AGL Personnel (Grove City, OH), and a Remedy Staffing CNC contract.
- Two-time Business Professionals of America Chapter 13 president; state and regional wins in Computer Programming, Team Mathematics, and Extemporaneous Speech.

VOLUNTEER WORK

- Systems administrator for a volunteer Linux community, digital radio station, and open shell environment.
- Long-term support for animal shelters, including events, outreach, and facilities work.
- OCR and digitization project contributor for historical legal documents.

TECHNICAL SKILLS

Languages: PHP, Perl, SQL, Bash, Python, Go, Rust, Zig, C#, TypeScript, C++ Web and Data: MySQL, MSSQL, PostgreSQL, WordPress, Redmine, Gitea, Wiki.js Infrastructure: Linux, Active Directory, Entra ID, Group Policy, PowerShell, Apache, Nginx, Caddy, IIS, Kubernetes, VMware ESXi, Datto backups, Cisco firewalls, Cisco Meraki, SonicWall, pfSense, OpenVPN, CrowdSec, Let's Encrypt/ACME, Cloudflare CDN Cloud: AWS, Azure, Google Cloud, Proxmox Security and Identity: Authentik, HashiCorp Vault, Psono, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, DMARC/SPF/DKIM, Mimecast, KnowBe4, DefectDojo, OpenVAS, zero-trust network design, incident response, CCPA/GDPR compliance work Collaboration: Agile, Scrum, Kanban, Redmine, Zammad, GitLab, CryptPad Mail Systems: iRedMail, Postfix, Dovecot Commerce: Shopify, custom ecommerce platforms, payment integrations Data and Automation: RAG design, OCR pipelines, semantic search, ETL, knowledge base design AI Agent Infrastructure: Agent Zero (A2A protocol, multi-node mesh, project isolation), Model Context Protocol (MCP server design), Composio toolkit integration, Venice AI SDK ecosystem (TypeScript/PHP SDKs), LiteLLM, multi-provider model routing across 13 authenticated providers (Anthropic, OpenAI, Z.AI GLM, MiniMax, Moonshot/Kimi, Venice, Airouter, Groq, Cerebras, Featherless, OpenRouter, NVIDIA, Mistral), tiered inference (utility/chat split), prompt caching, agent-driven CI and content pipelines, RAG and semantic search, agent security boundaries, scheduled autonomous tasks. Operational fluency across ten coding-agent CLIs (Claude Code, opencode, gemini-cli, Codex CLI, Kimi Code, OpenHands, Aider, Octofriend, Crush, letta-code) used as a benchmark and production stack. Systems and Networking: TCP/IP analysis, packet capture, PLC programming, industrial controls, prepress production (Esko, Prinergy, RIP workflows), IRC server administration Monitoring & Dashboards: Uptime Kuma, HostedScan, Red Sift OnDMARC, Dashy internal portal

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Currently studying: Stanford CS336, Language Modeling from Scratch (2025).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.